One-Way ANOVA

Statistical Methods Dossier with Subject-Specific Examples

1. Method Description

One-way ANOVA (Analysis of Variance) compares means of three or more groups.

The F-statistic = variance between groups / variance within groups.

A significant F-test indicates that at least one group mean differs from the others (omnibus test).

Post-hoc tests (Tukey HSD, Bonferroni, Scheffe) identify which specific groups differ.

Effect size: eta^2 = SS_between / SS_total. Partial eta^2 is used in more complex designs.

2. Subject-Specific Examples

Social Sciences: Do satisfaction levels differ across three departments (HR, Sales, IT)? DV: job satisfaction (1-7).

Economics: Do four investment strategies produce different average returns? Groups: strategy A, B, C, D. DV: annual return (%).

Human Sciences: Do three diets produce different weight loss? Groups: low-carb, low-fat, Mediterranean. DV: weight loss (kg).

Humanities: Do novels from three epochs differ in average sentence length? Groups: Romanticism, Realism, Modernism.

3. Important Concepts and Assumptions

Independence: Observations must be independent across groups.

Normality: DV should be normally distributed in each group (robust with n > 30 per group).

Homogeneity of variances: Variances equal across groups (Levenes test). Welch ANOVA as alternative.

Post-hoc multiplicity correction: Controls Type I error across multiple comparisons.

4. Interpretation and practice

Report: 'A one-way ANOVA revealed a significant effect of group on Y, F(3,96) = 5.67, p = .001, eta^2 = .15.'

Post-hoc: 'Tukey HSD post-hoc tests showed that Group A (M = 4.5, SD = 1.2) differed significantly from Group B (M = 3.1, SD = 1.1), p = .003.'

Planned contrasts: Test specific hypotheses instead of all pairwise comparisons (more powerful).

5. Code Examples

R: fit <- aov(y ~ group, data = d); summary(fit); TukeyHSD(fit)

Python: import statsmodels.api as sm; from statsmodels.formula.api import ols; fit = ols('y ~ C(group)', data=d).fit(); sm.stats.anova_lm(fit)

SPSS: ONEWAY y BY group /POSTHOC=TUKEY BONFERRONI /STATISTICS DESCRIPTIVES HOMOGENEITY.

R (nonparametric): kruskal.test(y ~ group, data = d)